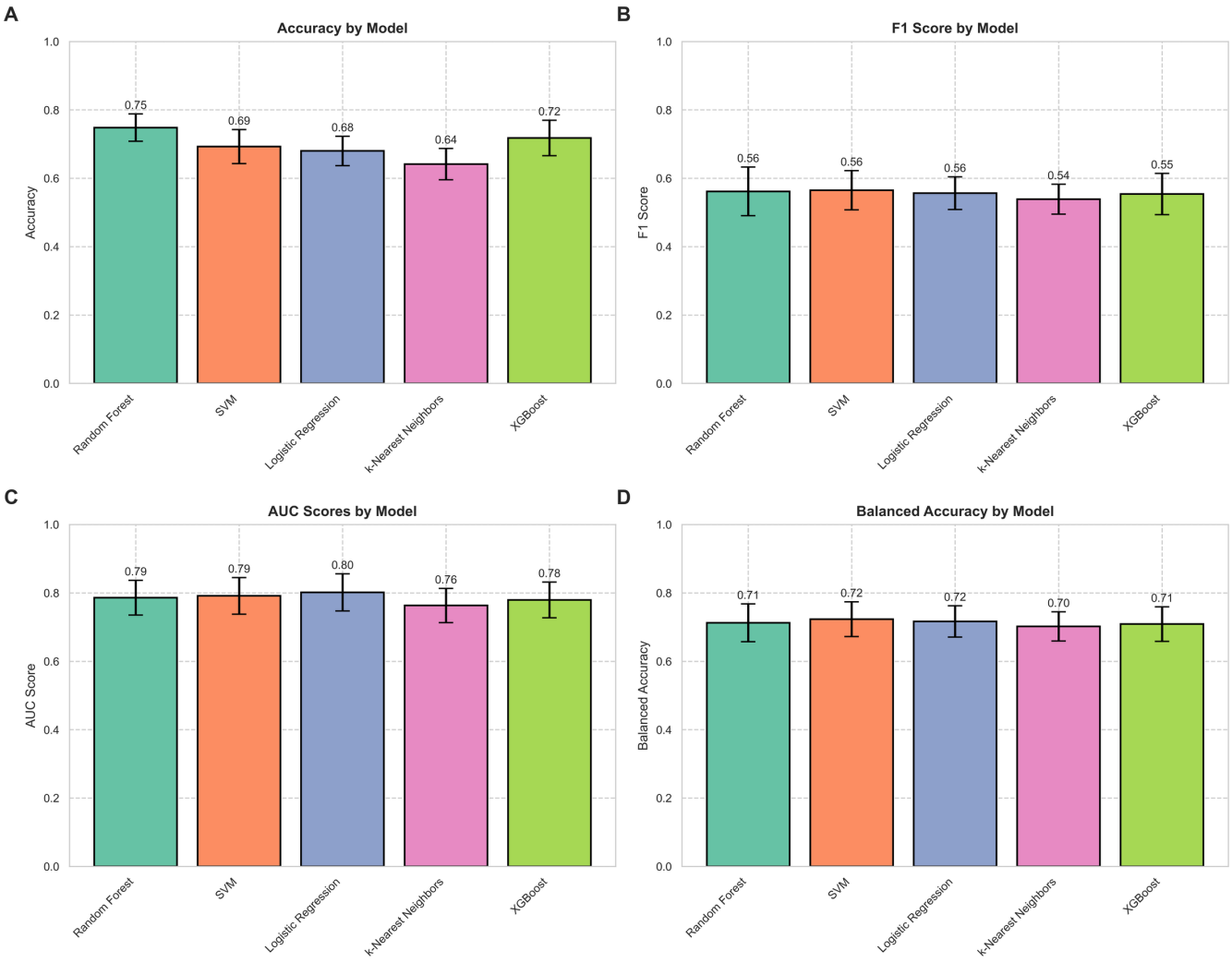
Supplementary Information

Uncovering novel pathogenic hotspots in the Pompe disease target using graph-based features and ensemble machine learning

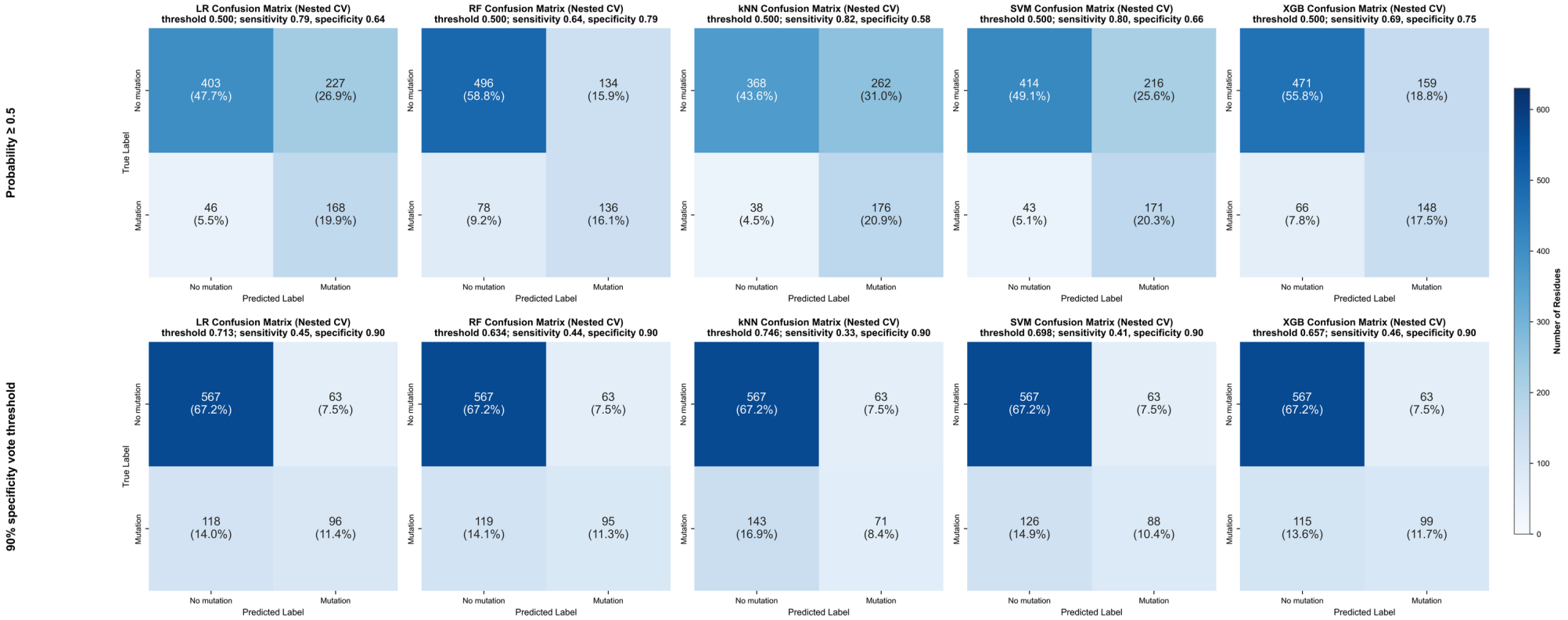
Sreeharsh K, Manjari Kiran, Mohd. Akif

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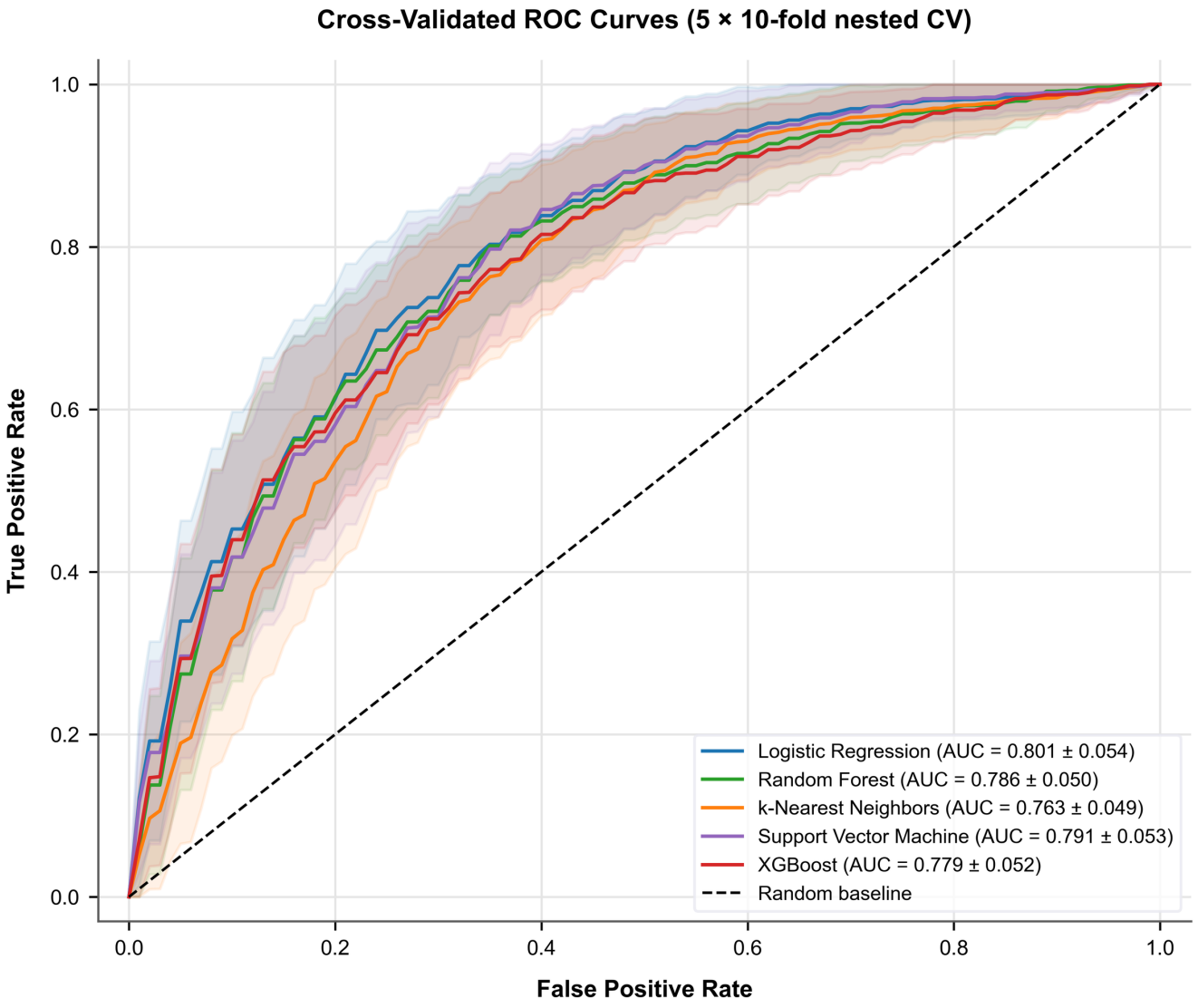
Supplementary Figure S1. Performance of the five classifiers in repeated nested cross-validation
Supplementary Figure S2. Confusion matrices of the five classifiers
Supplementary Figure S3. Cross-validated ROC curves
Supplementary Figure S4. Feature contributions to classifier predictions
Supplementary Figure S5. Stability of the novel consensus hotspots across cross-validation repeats
Supplementary Tables S1–S4 are provided as a separate Excel workbook (Supplementary_Tables.xlsx).



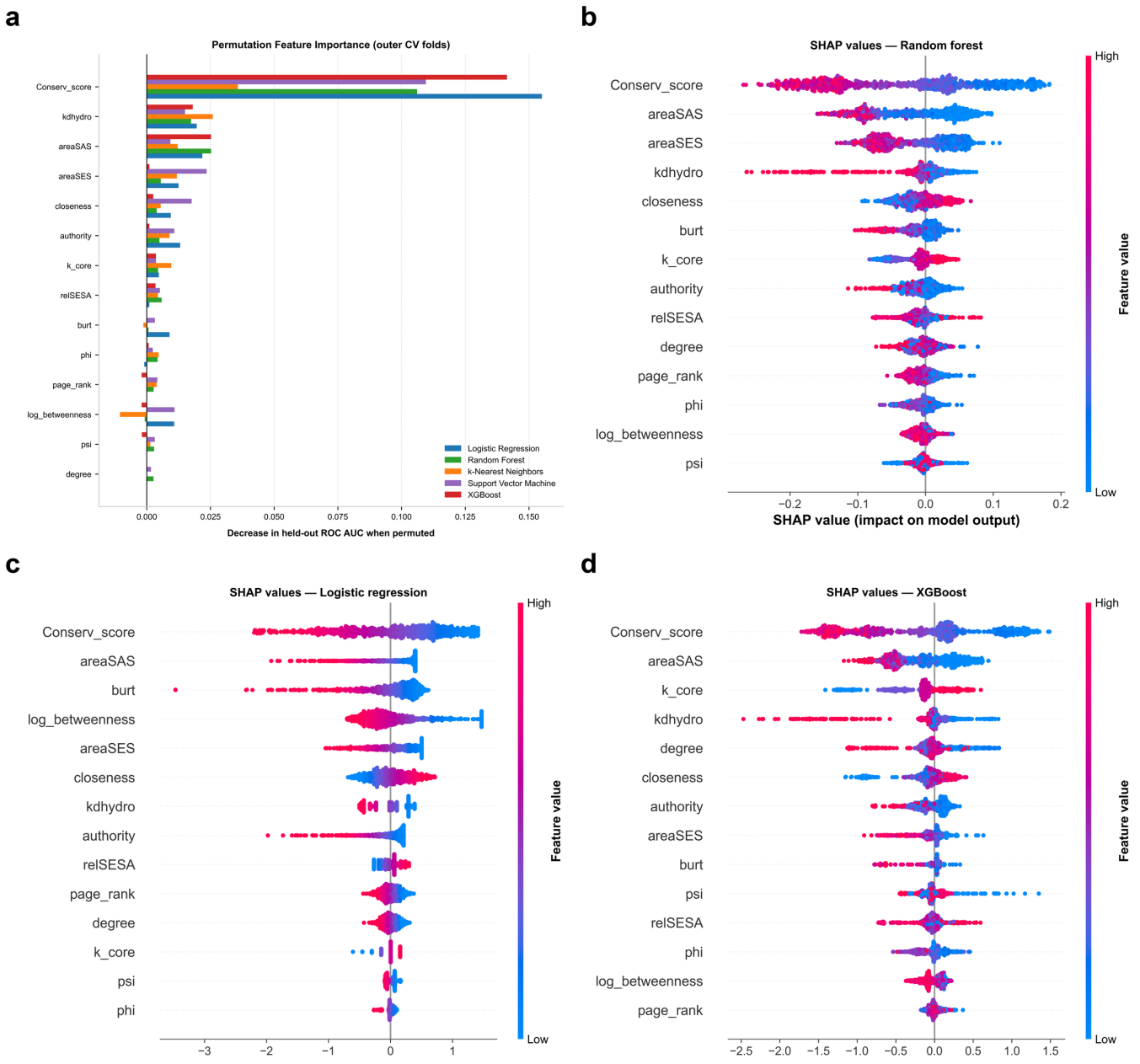
Supplementary Figure S1 | Performance of the five classifiers in repeated nested cross-validation. (a) Accuracy, (b) F1 score, (c) ROC AUC and (d) balanced accuracy of Random Forest, Support Vector Machine, Logistic Regression, k-nearest neighbors and XGBoost at a probability threshold of 0.5. Bars show the mean over the 50 outer test folds (5 repeats × 10 folds) and error bars the standard deviation. Fold-averaged AUCs are therefore slightly lower than the pooled out-of-fold AUCs reported in Fig. 4a.



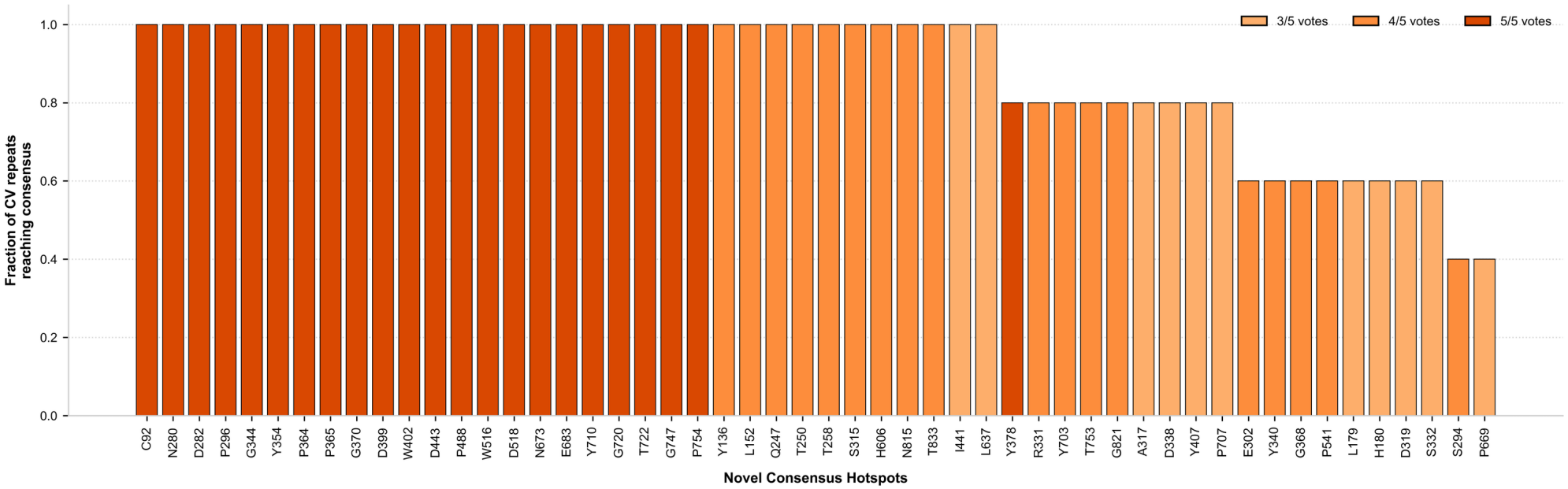
Supplementary Figure S2 | Confusion matrices of the five classifiers. Out-of-fold predictions for all 844 residues (214 mutation positions, 630 other residues). Top row: probability threshold of 0.5. Bottom row: model-specific threshold at 90% specificity, which was used for consensus voting. Each cell gives the number and percentage of residues; panel titles give the threshold, sensitivity and specificity. LR, Logistic Regression; RF, Random Forest; kNN, k-nearest neighbors; SVM, Support Vector Machine; XGB, XGBoost.



Supplementary Figure S3 | Cross-validated ROC curves. Mean ROC curve of each classifier over the 50 outer folds of repeated nested cross-validation; shaded bands show ±1 standard deviation. Legend values are the mean ± standard deviation of the fold AUCs. The dashed line represents random prediction.



Supplementary Figure S4 | Feature contributions to classifier predictions. (a) Permutation importance: decrease in held-out ROC AUC when each feature is randomly permuted in the outer test folds, shown for each classifier. (b–d) SHAP summary plots for Random Forest (b), Logistic Regression (c) and XGBoost (d) refitted on all residues with the selected hyperparameters; each point is one residue, coloured by its feature value. Conserv_score, ConSurf conservation score; kdhydro, Kyte–Doolittle hydrophobicity; areaSAS and areaSES, solvent-accessible and solvent-excluded surface area; relSESA, relative solvent-excluded surface area; burt, Burt's constraint; k_core, K-core; page_rank, PageRank-like score; phi and psi, backbone dihedral angles.



Supplementary Figure S5 | Stability of the novel consensus hotspots across cross-validation repeats. Fraction of the five cross-validation repeats in which each of the 52 novel consensus hotspots reached consensus (at least three of five classifier votes). Bars are coloured by the number of votes in the pooled analysis. Thirty-three positions reached consensus in all five repeats.